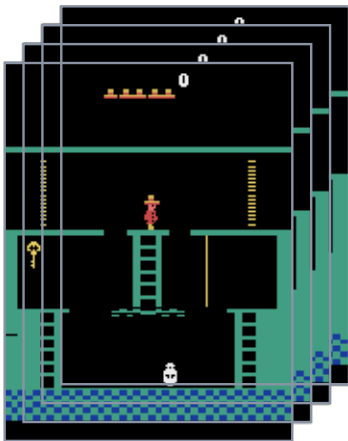
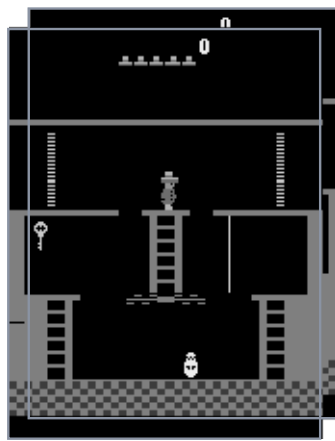


grayscale
(read frames 3 & 4)



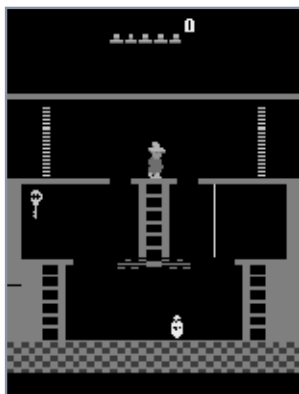
4 raw frames of one agent step
 $4 \times (210, 160, 3)$ RGB

np.maximum
(2 $\rightarrow$ 1)



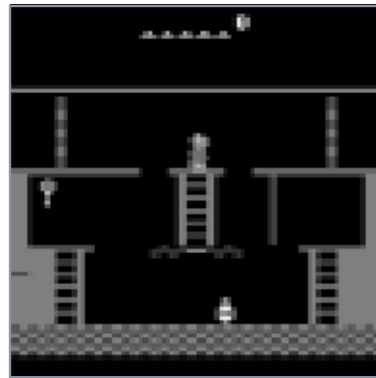
grayscale frames 3 & 4
 $2 \times (210, 160)$

resize
(INTER_AREA)



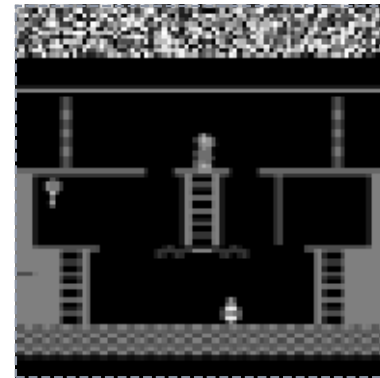
max-pooled
(210, 160)

+ noise patch
(TV runs only)

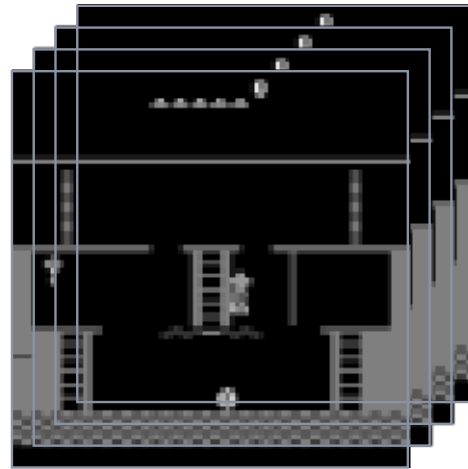


resized
(84, 84) uint8

push into deque
(1 frame / step)



noisy-TV stamped
(84, 84) — TV runs only



frame stack, last 4 agent steps
(4, 84, 84) uint8